

Critical swimming velocity and associated morphology of juvenile coastal cutthroat trout (*Oncorhynchus clarki clarki*), steelhead trout (*Oncorhynchus mykiss*), and their hybrids

D.K. Hawkins and T.P. Quinn

Abstract: Critical swimming velocities were determined for two hatchery stocks of cutthroat trout (*Oncorhynchus clarki clarki*), one hatchery stock of steelhead trout (*Oncorhynchus mykiss*), and reciprocal cutthroat–steelhead hybrids from both cutthroat stocks. Steelhead had the highest critical swimming velocity (7.69 body lengths/s), cutthroat had the lowest (5.58 and 6.69 body lengths/s for the two stocks), and hybrids were intermediate (7.14 and 7.22 body lengths/s). Consistent with the significant differences in swimming performance, principal components analysis revealed that cutthroat had shorter and narrower caudal regions than steelhead and that hybrids were morphologically intermediate. The morphology of hatchery populations compared with that of wild juvenile cutthroat and steelhead allowed us to determine if the interspecific morphological differences observed in hatchery-reared fish were representative of the species and did not result from hatchery practices, and to assess the validity of extrapolating the swimming performance results to wild fish. Although wild fish were generally more fusiform than hatchery fish of both species, the relative differences between cutthroat and steelhead shapes were the same for both hatchery-reared and wild fish. We conclude that hybrids are intermediate to the parental species in both swimming performance and morphology and that the potential for a competitive advantage exists.

Résumé : Nous avons calculé les vitesses critiques de nage pour deux stocks d'élevage de truite fardée (*Oncorhynchus clarki clarki*), un stock d'élevage de truite arc-en-ciel anadrome (*Oncorhynchus mykiss*) et des hybrides réciproques de ces deux truites provenant des deux stocks de truite fardée. La truite arc-en-ciel anadrome présentait la vitesse critique de nage la plus élevée (7,69 longueurs corporelles/s), la truite fardée la plus basse (5,58 et 6,69 longueurs corporelles/s pour les deux stocks), tandis que les hybrides donnaient des résultats intermédiaires (7,14 et 7,22 longueurs corporelles/s). Conformément aux différences significatives dans la performance de nage, l'analyse des composantes principales a révélé que la truite fardée présentait une région caudale plus courte et plus étroite que la truite arc-en-ciel anadrome, et que les hybrides possédaient des caractéristiques morphologiques intermédiaires. La comparaison de la morphologie des populations d'élevage à celle de juvéniles sauvages de truite fardée et de truite arc-en-ciel anadrome nous a permis de déterminer si les différences morphologiques interspécifiques observées chez les poissons d'élevage étaient représentatives de l'espèce et non pas dues aux pratiques piscicoles, et d'évaluer la validité d'une extrapolation aux poissons sauvages des résultats sur la performance de nage. Si les poissons sauvages étaient en général plus fusiformes que les spécimens d'élevage chez les deux espèces, les différences relatives entre la forme des deux espèces étaient les mêmes chez les poissons d'élevage et les poissons sauvages. Nous en concluons que les hybrides ont une position intermédiaire entre les espèces parentes tant pour la performance de nage que pour la morphologie, et qu'un avantage compétitif peut ainsi exister.

[Traduit par la Rédaction]

Introduction

Coastal cutthroat (*Oncorhynchus clarki clarki*) and steelhead trout (*Oncorhynchus mykiss*) coexist throughout much of their native ranges. Although much is yet to be learned about the timing of phyletic divergence (Utter and Allendorf 1994), Behnke (1992) concluded that cutthroat and steelhead shared a common ancestor as recently as 2 million years ago. As would be expected from this relatively recent divergence, cut-

throat and steelhead do not have many unique derived characteristics. However, differences at protein loci (Campton and Utter 1985; Leary et al. 1987), in mitochondrial DNA sequence (Wilson et al. 1985), in karyotype (Gold 1977), and in morphology (Hartman 1956; Stearley and Smith 1993) can be used to distinguish between the species.

Prior to 1981, hybridization between these two species was thought to be uncommon and had not been formally documented (Utter 1981; Verspoor 1988). However, Campton (1981) and Campton and Utter (1985) found a significant number of hybrids in two streams in the Puget Sound area of the state of Washington, indicating that hybridization occurs between coastal cutthroat and steelhead. Hybridization is cause for some concern because it may contribute to the decline of western trout species (Busack and Gall 1981), and many cutthroat populations currently are declining (Johnston and Mercer 1976; Trotter 1987; Nehlsen et al. 1991; Johnson et al. 1994).

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There are several reproductive isolating mechanisms, both pre- and post-mating, which maintain the species differences. Spawning adults are generally separated both temporally and spatially (Behnke 1972; Trotter 1987; Burgner et al. 1992). In western Washington, steelhead often spawn later in the season, using the deeper, faster water of larger rivers, whereas cutthroat spawn earlier in smaller headwaters (Johnston and Mercer 1976). Steelhead are also often much larger, which could lead to assortative mating or possibly to directionality in pair formation in the event of heterospecific matings (Foote and Larkin 1988). Furthermore, cutthroat and steelhead have different numbers of chromosomes (68 and 58–60, respectively; Behnke 1970), which could lead to mispairing of chromosomes and decreased viability in hybrid progeny.

Hybridization may be thought of as the breakdown of isolating mechanisms between two reproductively isolated species or populations (Mayr 1963). In fishes, hybridization is common within many families, owing in large part to the frequency of external fertilization and weak pair bonds (Hubbs 1955). There are many examples of hybridization among salmonids (Chevassus 1979; Paiute and Lahontan cutthroat, Busack and Gall 1981; golden and rainbow trout, Halliburton et al. 1983; arctic char and brook trout, Hammar et al. 1991; Apache, Gila, cutthroat, and rainbow trout, Dowling and Childs 1992; Atlantic salmon and brown trout, Jordan and Verspoor 1993). However, relative performance of the hybrid and parental offspring has seldom been compared in salmonids. In a review of the fitness of hybrids relative to their parents, Arnold and Hodges (1995) provided examples from many genera and concluded that hybrids are not uniformly less fit than the parental species. The relative fitness of the F_1 progeny determines whether occasional hybridization strengthens the isolating mechanisms or leads to introgression of the gene pools (Mayr 1963).

Sustained swimming ability is an important component of fitness because it affects a fish's habitat utilization, ability to perform extended migrations, and vulnerability to predation (Bams 1967; Tsuyuki and Williscroft 1977; Taylor and McPhail 1985; Bisson et al. 1988). Furthermore, swimming ability is correlated with body shape (Thomas and Donahoo 1977; Taylor and McPhail 1985). Taylor and Foote (1991) related differences in swimming ability among juvenile sockeye, kokanee, and their hybrids to differences in morphology and differences in the extent of adult migrations. Sockeye salmon, the anadromous form of *Oncorhynchus nerka*, undertake extensive migrations whereas kokanee do not. The sockeye body shape is consistent with the need for superior sustained swimming performance (longer, narrower body and longer caudal region). Taylor and McPhail (1985) showed similar differences between juvenile coastal and interior forms of coho salmon (*Oncorhynchus kisutch*). Interior coho, which undertake more extensive migrations as adults, are more fusiform and are better swimmers. Juvenile cutthroat and steelhead utilize different stream habitats; steelhead are generally found in waters with higher velocities (Thomas and Donahoo 1977; Bisson et al. 1988). There are also differences in the duration and extent of the adult marine migrations. Cutthroat generally spend only the summer in the ocean and are thought to remain near shore, whereas steelhead are at sea for 1 or more years and migrate long distances (Giger 1972; Trotter 1989; Burgner et al. 1992; Johnson et al. 1994). Consistent with these differ-

ences in juvenile habitat and adult migrations, steelhead are more cylindrical with short median fins and large paired fins useful for holding position in fast water (Hartman 1956; Bisson et al. 1988). Because of these differences in body shape, juvenile habitat preferences, and adult migrations between cutthroat and steelhead, we hypothesized that steelhead would be better swimmers than cutthroat.

The relative swimming performance of cutthroat, steelhead, and their hybrids is unknown. Reduced swimming ability could put hybrids at a competitive disadvantage compared with both cutthroat and steelhead, resulting in a barrier to gene flow. Enhanced swimming ability, on the other hand, could result in a competitive advantage for the hybrids, increasing the potential for introgression. The consequences of intermediate swimming ability of hybrids, relative to the parental species, would depend on the life-history pattern of the hybrids. Hybrids that adopt steelhead freshwater habitat and marine migrations could be competitively inferior because of selection for enhanced swimming ability in steelhead. However, hybrids that adopt cutthroat behaviors could be competitively superior to cutthroat. Bisson et al. (1988) described cutthroat as lacking morphological adaptations to either fast or slow water and suggested that this could be in part responsible for their displacement by other salmonids. If hybrids are able to attain higher critical swimming velocities than cutthroat, perhaps they could also displace cutthroat from more desirable habitat.

In this study, critical swimming velocities were determined for cutthroat, steelhead, and their hybrids to assess relative swimming abilities. Morphometric variation was also analyzed to determine if differences in swimming ability were related to differences in body shape. Hybrid performance and morphology were compared with the parental species to determine if cutthroat–steelhead hybrids were intermediate in both swimming ability and body shape. Finally, the variations in body morphology of the cutthroat and steelhead hatchery populations used in the swimming tests were compared with the body morphology of juvenile cutthroat and steelhead collected in the wild, as environmentally induced morphological differences between hatchery and wild fish could affect the ability to relate swimming performance differences determined with hatchery fish to wild populations (Taylor 1986; Swain et al. 1991).

Materials and methods

Adults from two coastal cutthroat trout stocks and one steelhead trout stock were used to create the progeny for this experiment. One stock of cutthroat and the steelhead were obtained from the Washington State Hatchery at Aberdeen, Washington. The other cutthroat stock came from the Washington State Hatchery at Shelton, Washington. The Lake Aberdeen cutthroat were identified as having some past genetic influence by subsequent protein electrophoresis, but seven of the eight adults used for gamete collection had only cutthroat trout alleles at all loci examined. Both stocks of cutthroat originated from anadromous fish, but neither stock is currently released. Brood stocks are maintained at the hatcheries and spawned as 3 year olds. The steelhead are released, and the hatchery stock is maintained using returning adults. Therefore, the steelhead undergo some selection for successful migration, whereas the cutthroat do not. Eggs and milt were obtained on 18 January (Aberdeen cutthroat and steelhead) and 27 January (Shelton cutthroat and Aberdeen steelhead), 1994, from

Fig. 1. Genetic design for cutthroat, steelhead, and hybrid progeny tested in swimming tunnels. In all fish types the dam is listed first and subscripts indicate origin of parent. C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton.

Aberdeen Hatchery (repeated four times, total 16 families)

	C _a C _a cutthroat dam	SS steelhead dam
C _a C _a cutthroat sire	C _a C _a pure cutthroat	S _a C _a paternal cutthroat hybrid
SS steelhead sire	C _a S _a maternal cutthroat hybrid	ESS early pure steelhead

Shelton Hatchery (repeated 6 times, total 24 families)

	C _s C _s cutthroat dam	SS steelhead dam
C _s C _s cutthroat sire	C _s C _s pure cutthroat	S _a C _s paternal cutthroat hybrid
SS steelhead sire	C _s S _a maternal cutthroat hybrid	LSS late pure steelhead

the two hatcheries and transported to the hatchery at the University of Washington, School of Fisheries, Seattle. The eggs from each female were divided into two lots, one fertilized by a conspecific male and one fertilized by a heterospecific male. The same male was used to fertilize one conspecific and one heterospecific lot of eggs. Each pair of cutthroat and steelhead females shared a pair of cutthroat and steelhead males so that the hybrid families created by a pair of females were maternal or paternal half-sibs to the pure cutthroat and steelhead families. Ten pairs of steelhead, four pairs of Aberdeen cutthroat, and six pairs of Shelton cutthroat were used, resulting in 40 families, 10 of each major cross type. The four major cross types were CC, pure cutthroat; CS, hybrids with cutthroat dams; SC, hybrids with steelhead dams; and SS, pure steelhead. Because the Shelton cutthroat were spawned 1 week later than the Aberdeen cutthroat, the steelhead used in creating the Shelton families were also collected 1 week later. Therefore, each major cross type had two minor cross types that could be compared. The eight minor cross types were C_aC_a, Aberdeen cutthroat; C_sC_s, Shelton cutthroat; C_aS_a, hybrids with Aberdeen cutthroat dams; C_sS_a, hybrids with Shelton cutthroat dams; S_aC_a, hybrids with Aberdeen cutthroat sires; S_aC_s, hybrids with Shelton cutthroat sires; ESS, early Aberdeen steelhead used in crosses with Aberdeen cutthroat; and LSS, late Aberdeen steelhead used in crosses with Shelton cutthroat (Fig. 1).

Families were incubated separately in randomly assigned Heath trays using dechlorinated city water maintained at 10 ± 1°C. As each family reached the swim-up stage in late March, they were transferred to 19-L rearing buckets supplied with Lake Washington water, which warmed from 10 to 19°C as the season progressed. The fish were fed Moore–Clark salmon diet three times daily for 1 month and once daily thereafter to satiation. Throughout the rearing and testing periods the fish experienced controlled lighting (12 h light : 12 h dark) to prevent changes in morphology caused by exogenous day-length cues. Because of heavy mortalities in both pure Aberdeen cutthroat

and pure Shelton cutthroat, additional cutthroat were obtained from the hatcheries as fry on 31 May and 13 June, respectively. These fry were a mixture of the progeny produced at the hatchery that year and therefore represented many families. At the School of Fisheries hatchery these fish were maintained in separate 19-L containers in the same water and under the same husbandry regimen as the initial crosses until testing.

Critical swimming velocities were determined in a modified Blazka-type swim chamber (Smith and Newcomb 1970; Flagg 1981; Moser et al. 1990) consisting of a Plexiglas swimming tunnel (16 cm diameter, 71 cm long) contained within another larger Plexiglas tube (24 cm diameter). Water flow was generated by an impeller at the rear of the chamber controlled by a variable-speed, 1-hp (1 hp = 745.70 W) DC motor. Another impeller at the front of the chamber and a bundle of plastic straws inserted into the chamber reduced the friction at the side of the chamber and created a more equally distributed flow (Carpenter 1987). An electrified screen placed before the rear impeller kept the fish from contacting the propeller or leaving the testing chamber. The electric current to the screen, controlled by a rheostat, prevented the fish from resting on the screen.

Fish were tested at 4–5 months of age (9 July – 29 August), in lake water at the same temperature as their rearing water. Testing occurred from 0.5 h after lights on until 0.5 h before lights off. On each day, fish from the different types were tested in a random order to minimize any effect of time of day on swimming performance. For each test a fish was randomly selected from the rearing bucket. The fork length was estimated by placing the fish in a small, water-filled plastic tube containing a ruler before it was placed in the swimming tunnel. The estimated fork length was used to calculate water velocities in body lengths per second (bl/s) to avoid exposing the fish to anesthetic or excessive handling prior to testing. Basing water velocities on fork length allowed comparison among fish of different sizes because swimming ability has been shown to be related to size (Brett 1965; Thomas and Donahoo 1977). After a 5-min acclimation period, the water velocity was gradually increased from zero to the initial velocity of 3 bl/s over a 5-min period, then maintained at this speed for 15 min. Each 15 min thereafter, the velocity was increased by 1 bl/s over 1 min. This procedure was continued until the fish was no longer able to maintain position and fell against the electrified screen. At this point, the electricity and water current were turned off, the time elapsed in the last 15-min period was recorded, and the fish was removed from the tunnel.

After the test, the fish was anaesthetized with MS-222 buffered with sodium bicarbonate, morphological measurements were taken, the adipose fin was clipped to prevent retesting, and the fish was returned to the rearing bucket. The regression relationship between the estimated fork length and the actual fork length indicated that the estimated lengths were accurate and unbiased over the size range of fish tested ($r^2 = 0.99$, $n = 133$, slope = 0.96, y intercept = 3.33 mm). Therefore, swimming velocities were not corrected for differences between the two measurements.

Critical swimming velocity (U_{crit} ; body lengths per second) was calculated using the formula given by Brett (1964):

$$U_{crit} = V_p + (T_f/T_i)V_i$$

where V_p is the velocity at which the fish last swam for the full 15-min period, V_i is the velocity increment (1 bl/s), T_f is the elapsed time from the last velocity increase to fatigue, and T_i is the time between velocity increases (15 min).

We anticipated that cutthroat and steelhead reared under common conditions would display species-specific morphological differences and that these might be associated with patterns of swimming performance. However, the performance and morphology of the hybrids could resemble one or the other parental species or be different from either. Therefore, we measured the morphology of all fish tested to relate body shape and swimming performance to each other and to the parents.

Table 1. Sample sizes and mean critical swimming velocities (bl/s, mean $\pm$ SE) for groups tested in the swimming tunnel, prior to pooling between minor cross types.

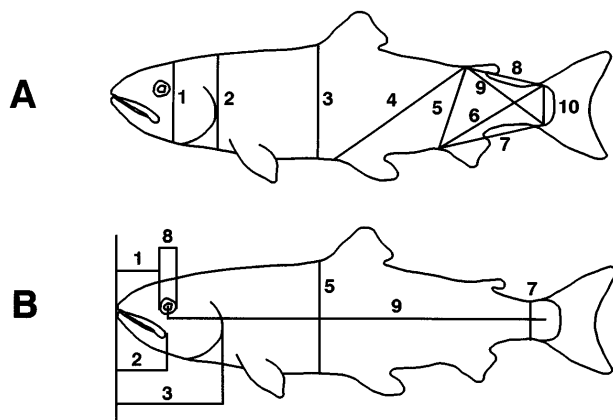
Fish type	No. tested	No. of families	Mean critical swimming velocity ^a
C _a C _a Aberdeen cutthroat received as eggs	2	2	6.38 $\pm$ 0.96 (1)
C _a C _a Aberdeen cutthroat received as fry	11	Unknown ^b	6.74 $\pm$ 0.23 (1)
C _s C _s Shelton cutthroat received as eggs	12	3	5.71 $\pm$ 0.22 (2)
C _s C _s Shelton cutthroat received as fry	11	Unknown ^b	5.45 $\pm$ 0.20 (2)
C _a S _a maternal Aberdeen cutthroat hybrid	9	3	7.29 $\pm$ 0.27 (3)
C _s S _a maternal Shelton cutthroat hybrid	22	3	7.08 $\pm$ 0.18 (3)
S _a C _a paternal Aberdeen cutthroat hybrid	15	3	7.40 $\pm$ 0.24 (4)
S _a C _s paternal Shelton cutthroat hybrid	20	4	7.09 $\pm$ 0.12 (4)
ESS early spawned Aberdeen steelhead	16	3	7.50 $\pm$ 0.27 (5)
LSS late spawned Aberdeen steelhead	15	3	7.90 $\pm$ 0.18 (5)
Total	133	24 ^b	na

Note: In all fish types the dam is listed first and subscripts indicate origin of parent: C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton. na, not applicable.

^aValues in parentheses indicate groups pooled on the basis of *t* tests.

^bCutthroat received as fry were from a large raceway containing mixed fry from the fish spawned at the hatchery that year.

Fig. 2. (A) Morphological traits used in principal components analysis of fish tested in swimming tunnels. 1, head depth 1 (HD1); 2, head depth 2 (HD2); 3, body depth at insertion of dorsal fin (PELVDF); 4, insertion of pelvic fin to insertion of adipose fin (PELVAD); 5, insertion of anal fin to insertion of adipose fin (ANAD); 6, insertion of anal fin to top of caudal flexure (ANC1); 7, insertion of anal fin to bottom of caudal flexure (ANC2); 8, insertion of adipose fin to top of caudal flexure (ADC1); 9, insertion of adipose fin to bottom of caudal flexure (ADC2); 10, caudal flexure depth (CFD). (B) Morphological traits used in principal components analysis of hatchery and wild cutthroat and steelhead. 1, snout; 2, maxillary length; 3, head length; 4, mouth width (see Hartman 1956); 5, body depth; 6, weight (not shown); 7, caudal depth; 8, eye diameter; 9, mid-eye hypural length (MEH).



We conducted the swimming tests on fish reared under controlled conditions to prevent differential experience from biasing the results. For example, wild steelhead might outperform cutthroat in part because they rear in faster water. However, to relate our findings to wild fish, we compared the morphology of the experimental fish with that of wild fish of both species.

The morphological measurements taken on the 133 fish tested in the swimming tunnel (Table 1) were the modified truss measurements used by Taylor and Foote (1991) to compare overall body depth, body length, and depth of the caudal region used in swimming (Fig. 2A). In addition, 89 wild coastal cutthroat and 26 wild steelhead trout were collected from Mission Creek (near Belfair, Wash.) and Harvey Creek

(near Arlington, Wash.) in September 1994. Morphological measurements useful in differentiating between juvenile cutthroat and steelhead in the field were taken on these fish. Although these measures did not include all of those used in the comparison of the fish tested in the swimming tunnel, they included measures of body length, body depth, and caudal depth (Fig. 2B). The hatchery-reared fish tested in the swimming tunnel were subjected to both sets of morphologic measurements to facilitate comparison with the wild fish.

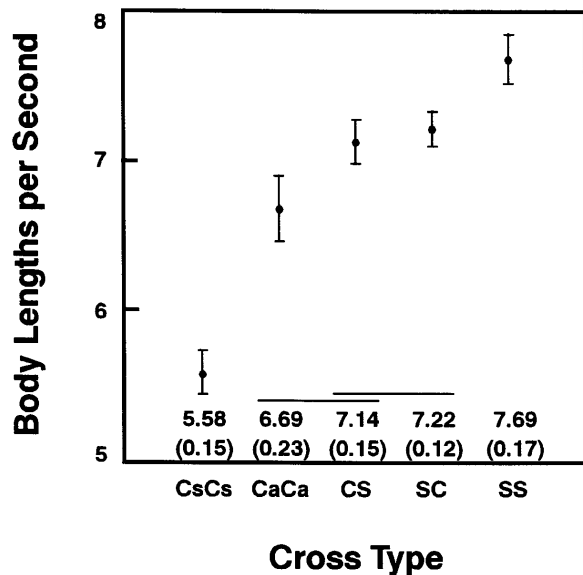
The primary focus of this study was to compare the swimming performance of cutthroat, steelhead, and reciprocal hybrids. However, we considered that there might be population-specific differences so two stocks of cutthroat were used, creating eight minor cross types. Also, because of the early cutthroat mortalities, each type of pure cutthroat had two subtypes, those received as eggs and those received as fry. To determine whether we could pool minor cross types within a major cross type, and pool the subtypes of pure cutthroat (i.e., were Aberdeen and Shelton cutthroat the same, were early and late steelhead the same, were all maternal cutthroat hybrids the same regardless of the dam's stock of origin, were all paternal cutthroat hybrids the same regardless of the sire's stock of origin, and were all pure Aberdeen cutthroat or pure Shelton cutthroat the same regardless of when they were obtained), the mean critical swimming velocities were compared using two sample *t* tests. After pooling types based on the *t* tests, mean critical velocities were compared among the remaining cross types using analysis of variance (ANOVA) followed by Fisher's least significant difference test. Fisher's test compares the ranked means of adjacent pairs with the minimum difference necessary to declare significant differences (Zar 1984). Correlations between swimming ability and fork length, and swimming ability and date of test, were also examined using Spearman's rank correlation coefficient.

Principal components analysis with varimax rotation (Tabachnick and Fidell 1989) was used to assess body shape differences associated with swimming performance and to compare the hatchery-reared populations with those collected in the wild. Character component loadings indicate which characters are combined in each principal component. Varimax rotation rotates the component axes to maximize the variation between each principal component, thus enhancing interpretation of the data. To correct for allometric growth, all morphologic measurements were standardized to the average fork length prior to principal components analysis using the adjustment equation suggested by Ihssen et al. (1981) with Reist's (1986) adjustment coefficient:

$$M_{\text{adj}} = M_o(L/L_o)^b$$

where M_{adj} is the standardized measurement, M_o is the observed meas-

Fig. 3. Mean critical swimming velocities (bl/s) with standard error bars for pooled cross types. Values and results of Fisher's least significant difference test are given at the bottom of the graph.



urement, L is the mean fork length of the group of fish compared, L_o is the observed fork length, and b is the adjustment coefficient. The adjustment coefficient is the slope of the regression of $\log_{10}$ observed measurement on $\log_{10}$ fork length and is calculated separately for each morphological measurement. Differences in mean principal component scores among the fish types were analyzed using ANOVA followed by Tukey's test. Tukey's test was used because it allows comparisons among all groups.

Results

Swimming performance

The mean critical swimming velocities and the results of the t tests used to pool groups prior to ANOVA are given in Table 1. There were no differences in critical swimming velocities between cutthroat obtained as eggs and those obtained as fry from either hatchery (Aberdeen: $t = -0.555$, $df = 11$, $p = 0.59$; Shelton: $t = 0.236$, $df = 19$, $p = 0.81$), or between early-spawned and late-spawned steelhead ($t = -1.216$, $df = 29$, $p = 0.23$). Likewise, no between-population differences were found in cutthroat hybrids (maternal cutthroat: $t = 0.630$, $df = 29$, $p = 0.53$; paternal cutthroat: $t = 1.144$, $df = 21$, $p = 0.26$). However, critical swimming velocities between the two pure stocks of cutthroat differed ($t = 4.775$, $df = 32$, $p < 0.001$); Aberdeen cutthroat outperformed Shelton cutthroat by slightly more than 1 bl/s.

The t -test comparisons resulted in five types of fish to be tested using ANOVA: Aberdeen cutthroat (C_aC_a), Shelton cutthroat (C_sC_s), maternal cutthroat hybrids (CS), maternal steelhead hybrids (SC), and steelhead (SS). The variances of these five types were similar (Bartlett's test, $p > 0.5$). The mean critical velocities of the five types of fish differed ($F = 24.397$, $df = 4, 128$, $p < 0.001$). Shelton cutthroat were the poorest swimmers ($U_{crit} = 5.58 \pm 0.15$ bl/s; mean $\pm$ SE), Aberdeen cutthroat were slightly better swimmers ($U_{crit} = 6.69 \pm 0.23$ bl/s), steelhead juveniles performed the best ($U_{crit} = 7.69 \pm 0.17$ bl/s), and hybrids were intermediate in swimming ability (CS,

Table 2. Fork length (FL (mm), mean $\pm$ SD) for all fish types used in swimming tests and Spearman's rank correlation of critical swimming velocity (bl/s) with fork length and date of test, after pooling between minor cross types.

Fish type	Sample size (n)	FL	Correlation	
			U_{crit} vs. FL	U_{crit} vs. date
C_aC_a	13	88.6 $\pm$ 7.7	-0.421	0.352
C_sC_s	23	89.6 $\pm$ 12.0	-0.420*	-0.477*
CS	31	87.6 $\pm$ 11.0	-0.512***	-0.513**
SC	35	87.3 $\pm$ 13.0	-0.444**	-0.197
SS	31	100.1 $\pm$ 9.9	0.017	-0.091
Total	133	90.9 $\pm$ 12.2	-0.130	-0.154

Note: In all fish types the dam is listed first and subscripts indicate origin of parent: C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.005$.

$U_{crit} = 7.14 \pm 0.15$ bl/s; SC, $U_{crit} = 7.22 \pm 0.12$ bl/s). Significant differences were identified using Fisher's least significant difference test ($df = 128$). No differences were found between maternal and paternal cutthroat hybrids ($p = 0.69$) nor between Aberdeen cutthroat and maternal cutthroat hybrids ($p = 0.095$). Shelton cutthroat were significantly different from Aberdeen cutthroat ($p < 0.001$) and steelhead were significantly different from paternal cutthroat hybrids ($p = 0.02$) (Fig. 3).

There was a large range in fork length both within and among test groups (Table 2). Although the mean fork lengths differed among groups ($F = 7.002$, $df = 4, 128$, $p < 0.001$), only the steelhead were different from the other groups (Tukey, $p < 0.05$ for all comparisons of steelhead with other test groups). Therefore, Spearman's r was calculated to test for correlation between fork length and swimming performance. The values ranged from -0.512 (CS; $p < 0.005$) to 0.017 (SS; $p > 0.50$) and were negative for all groups except steelhead. The correlation for all of the fish pooled was -0.130 ($p > 0.10$), indicating a slight tendency for smaller fish to swim faster in terms of relative speed (body lengths per second) (Table 2). Correlations between date of testing and swimming performance were also calculated because the testing period spanned 2 months. The values of Spearman's r ranged from -0.513 (CS; $p < 0.005$) to 0.352 (C_aC_a ; $p > 0.20$) with the value for all of the fish pooled (-0.154, $p > 0.05$) indicating a slight tendency for fish tested early to perform better (Table 2).

Morphology

The standard errors for all of the morphological measurements taken were generally small, indicating little variation within forms (Table 3). The exception was the higher standard errors associated with the measurement of length, which is a result of the large range in size of the fish.

All measurements were size adjusted to the common fork length of the groups compared (hatchery vs. wild mean fork length = 83.91 mm, swimming tunnel fish mean fork length = 90.86 mm), so that the principal components extracted described variation in the shape of the fish. This conclusion is supported by the low values of Spearman's r calculating correlations between the principal components and fork length (Table 4).

Table 3. Morphological measurements (mean $\pm$ SE, not size adjusted) used in the principal components analyses for each type of fish measured.

	Fish type								
	C _w C _w	S _w S _w	C _a C _a	C _s C _s	C _a S _a	C _s S _a	S _a C _a	S _a C _s	SS
Sample size	89	26	13	23	9	22	15	20	31
Fork length	75.76 $\pm$ 3.18	77.13 $\pm$ 6.85	88.57 $\pm$ 2.13	89.62 $\pm$ 2.50	94.69 $\pm$ 2.92	84.64 $\pm$ 2.28	90.42 $\pm$ 3.24	84.90 $\pm$ 2.94	100.10 $\pm$ 1.77
Snout	4.39 $\pm$ 0.21	4.22 $\pm$ 0.34	4.86 $\pm$ 0.18	5.11 $\pm$ 0.17	4.86 $\pm$ 0.24	4.69 $\pm$ 0.15	4.84 $\pm$ 0.17	4.59 $\pm$ 0.19	4.92 $\pm$ 0.12
Maxillary	8.73 $\pm$ 0.41	7.84 $\pm$ 0.61	9.09 $\pm$ 0.22	9.55 $\pm$ 0.28	9.28 $\pm$ 0.25	8.54 $\pm$ 0.17	8.89 $\pm$ 0.23	8.88 $\pm$ 0.24	9.32 $\pm$ 0.16
Head length	18.46 $\pm$ 0.77	17.89 $\pm$ 1.51	19.35 $\pm$ 0.42	20.33 $\pm$ 0.54	20.51 $\pm$ 0.57	18.77 $\pm$ 0.50	19.88 $\pm$ 0.56	18.97 $\pm$ 0.59	21.23 $\pm$ 0.32
Mouth width	6.61 $\pm$ 0.27	6.19 $\pm$ 0.49	7.69 $\pm$ 0.25	8.25 $\pm$ 0.24	8.17 $\pm$ 0.25	7.70 $\pm$ 0.16	7.85 $\pm$ 0.29	7.76 $\pm$ 0.24	8.56 $\pm$ 0.14
Body depth	14.87 $\pm$ 0.61	16.08 $\pm$ 1.40	18.08 $\pm$ 0.47	19.21 $\pm$ 0.51	19.71 $\pm$ 0.72	17.78 $\pm$ 0.61	18.50 $\pm$ 0.76	17.58 $\pm$ 0.65	20.89 $\pm$ 0.46
Weight	6.72 $\pm$ 1.09	8.34 $\pm$ 2.32	8.40 $\pm$ 0.63	9.44 $\pm$ 0.70	10.31 $\pm$ 0.95	8.18 $\pm$ 0.76	9.28 $\pm$ 0.96	7.94 $\pm$ 0.80	12.70 $\pm$ 0.69
Caudal depth	6.39 $\pm$ 0.25	6.60 $\pm$ 0.57	7.56 $\pm$ 0.20	7.47 $\pm$ 0.18	8.41 $\pm$ 0.28	7.06 $\pm$ 0.18	7.61 $\pm$ 0.30	7.20 $\pm$ 0.24	8.54 $\pm$ 0.17
Eye diameter	5.25 $\pm$ 0.18	5.51 $\pm$ 0.34	5.49 $\pm$ 0.14	5.45 $\pm$ 0.11	6.06 $\pm$ 0.11	5.58 $\pm$ 0.10	5.84 $\pm$ 0.17	5.79 $\pm$ 0.17	6.36 $\pm$ 0.09
Mid-eye-hypural	62.28 $\pm$ 2.67	64.24 $\pm$ 5.87	74.24 $\pm$ 1.90	74.89 $\pm$ 2.27	79.01 $\pm$ 2.70	70.76 $\pm$ 1.98	76.15 $\pm$ 2.86	70.67 $\pm$ 2.65	84.57 $\pm$ 1.71
HD1			12.58 $\pm$ 0.30	13.64 $\pm$ 0.40	13.69 $\pm$ 0.42	12.39 $\pm$ 0.31	13.12 $\pm$ 0.46	12.60 $\pm$ 0.48	14.21 $\pm$ 0.28
HD2			15.88 $\pm$ 0.43	16.90 $\pm$ 0.44	17.21 $\pm$ 0.59	15.64 $\pm$ 0.48	16.23 $\pm$ 0.58	15.66 $\pm$ 0.54	17.91 $\pm$ 0.36
PELVDF			17.80 $\pm$ 0.55	18.66 $\pm$ 0.47	19.50 $\pm$ 0.59	17.04 $\pm$ 0.56	18.39 $\pm$ 0.78	17.08 $\pm$ 0.65	20.78 $\pm$ 0.47
PELVAD			26.12 $\pm$ 0.78	26.31 $\pm$ 0.85	27.00 $\pm$ 1.23	25.21 $\pm$ 0.86	25.67 $\pm$ 1.43	24.60 $\pm$ 0.88	29.38 $\pm$ 0.68
ANAD			13.24 $\pm$ 0.35	13.70 $\pm$ 0.40	15.19 $\pm$ 0.44	13.42 $\pm$ 0.53	14.06 $\pm$ 0.68	13.10 $\pm$ 0.51	15.86 $\pm$ 0.36
ANC1			21.98 $\pm$ 0.77	22.02 $\pm$ 0.67	23.81 $\pm$ 0.57	20.78 $\pm$ 0.58	22.21 $\pm$ 0.92	20.82 $\pm$ 0.86	24.80 $\pm$ 0.47
ANC2			18.99 $\pm$ 0.50	19.91 $\pm$ 0.60	20.92 $\pm$ 0.45	18.89 $\pm$ 0.52	19.84 $\pm$ 0.96	19.24 $\pm$ 0.72	22.27 $\pm$ 0.44
ADC1			13.97 $\pm$ 0.47	13.60 $\pm$ 0.40	15.84 $\pm$ 0.50	13.07 $\pm$ 0.37	14.51 $\pm$ 0.53	13.12 $\pm$ 0.50	16.10 $\pm$ 0.34
ADC2			15.87 $\pm$ 0.44	16.08 $\pm$ 0.47	17.70 $\pm$ 0.52	15.16 $\pm$ 0.38	16.58 $\pm$ 0.68	15.41 $\pm$ 0.64	18.85 $\pm$ 0.35
CFD			7.07 $\pm$ 0.21	6.83 $\pm$ 0.18	7.75 $\pm$ 0.25	6.75 $\pm$ 0.16	7.28 $\pm$ 0.26	6.90 $\pm$ 0.20	8.10 $\pm$ 0.18

Note: The truss measurements (see Fig. 2A) were not taken on the wild fish. All measurements are in millimetres except weight, which is in grams.

Table 4. Spearman's rank correlation of principal components with fork length (after size adjustment of traits).

Comparison	PC1	PC2	PC3	PC4
Hatchery–wild	–0.178*	0.294***	–0.037	–0.020
Swimming fish	–0.041	0.234**	–0.065	0.056

Note: Hatchery–wild indicates comparison of hatchery-reared cutthroat and steelhead with wild cutthroat and steelhead. Swimming fish indicates comparison of the five major groups tested in the swimming tunnel.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

In the comparison of morphology among the seven types of fish tested in the swimming tunnel (C_aC_a, C_sC_s, C_aS_a, C_sS_a, S_aC_a, S_aC_s, and SS) using the modified truss measurements, four principal components (PC1–PC4) accounted for 68.58% of the variation. In general, PC1 summarized the depth of the fish forward of the dorsal fin, PC2 summarized the length of the caudal region, PC3 summarized the length of the fish behind the dorsal fin, and PC4 summarized the depth of the caudal region. The rotated component loadings and the mean component scores for each type of fish tested are shown in Table 5. PC2 (caudal region length) and PC4 (caudal region depth) were analyzed to compare the seven types because this part of the body is used for propulsion (Taylor and McPhail 1985). These components varied among types (PC2: $F = 4.92$, $df = 6, 126$, $p < 0.001$; PC4: $F = 3.03$, $df = 6, 126$, $p = 0.008$). In general, both Aberdeen and Shelton cutthroat had narrower caudal regions than steelhead or any of the hybrid types, although only steelhead and Shelton cutthroat differed significantly using Tukey's tests. Hybrids, with the exception of

maternal Aberdeen cutthroat, tended to be intermediate (Fig. 4). When length of the caudal region was analyzed, Aberdeen cutthroat and all fish with Shelton cutthroat parents (C_sC_s, C_sS_a, and S_aC_s) had shorter caudal regions. Maternal Aberdeen cutthroat were again the hybrid exception, with longer caudal regions than any other type. Tukey's tests indicated significant differences between both steelhead and hybrids with Aberdeen cutthroat mothers and all fish with Shelton cutthroat parents (Fig. 4). Additionally, there was a positive correlation (Spearman's $r = 0.227$, $n = 133$, $p < 0.01$) between critical swimming velocity and size-adjusted caudal depth, indicating that fish with deeper caudals for their size attained higher critical swimming velocities.

In the comparison of the cutthroat and steelhead hatchery populations with wild fish using the field identification measurements, four principal components (PC1–PC4) accounted for 79.67% of the morphological variation. PC1 summarized variation in head size, PC2 summarized body and caudal depth, PC3 summarized eye size, and PC4 summarized mid-eye-hypural (MEH) length. The rotated component loadings as well as the mean component scores for the four types of fish are shown in Table 6. PC3 was not used in the comparison of hatchery and wild fish because eye size was not expected to affect swimming ability. The wild fish were generally more fusiform than the hatchery fish. Figure 5 depicts the relationships among the five groups along PC1, PC2, and PC4. In general, although the hatchery populations tended to group together, separately from the wild populations, the relationship between cutthroat and steelhead within a rearing type was consistent. Cutthroat had longer and wider heads, longer MEH measurements, and slimmer bodies than steelhead. ANOVA detected differences in each of the three principal components

Table 5. Rotated character component loadings (used to determine which characters comprise each principal component) and group mean component scores for the principal components analysis of morphological variation comparing the seven groups of cutthroat, hybrids, and steelhead tested in the swimming tunnel (see Fig. 2A for definition of characters).

(A) Rotated component loadings					
Character	Fig. 2A No.	Rotated component loadings			
		PC1	PC2	PC3	PC4
HD2	2	0.898	-0.079	0.113	0.066
PELVDF	3	0.807	0.169	0.018	0.169
HD1	1	0.759	-0.252	0.051	-0.134
ADC1	8	-0.088	0.864	0.023	-0.111
ADC2	9	-0.070	0.797	-0.044	0.259
PELVAD	4	0.050	-0.314	0.772	0.016
ANC2	7	0.076	0.384	0.727	0.026
CFD	10	-0.040	0.026	-0.097	0.901
ANAD	5	0.223	0.093	0.392	0.564
ANC1	6	0.040	0.495	0.447	0.024
% variation		21.08	19.82	15.06	12.62

(B) Mean component scores				
Group	Mean component scores			
	PC1	PC2	PC3	PC4
C _a C _a	-0.548	-0.270	-0.092	-0.226
C _s C _s	0.862	-0.345	0.085	-0.667
C _a S _a	0.005	0.842	-0.312	0.391
C _s S _a	-0.056	-0.445	0.541	0.053
S _a C _a	-0.175	0.216	-0.565	0.023
S _a C _s	0.039	-0.282	0.260	0.100
SS	-0.312	0.519	-0.212	0.363

Note: The percentage of total variation explained by each component is given below the character component loadings. In all fish types the dam is listed first and subscripts indicate origin of parent. C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton.

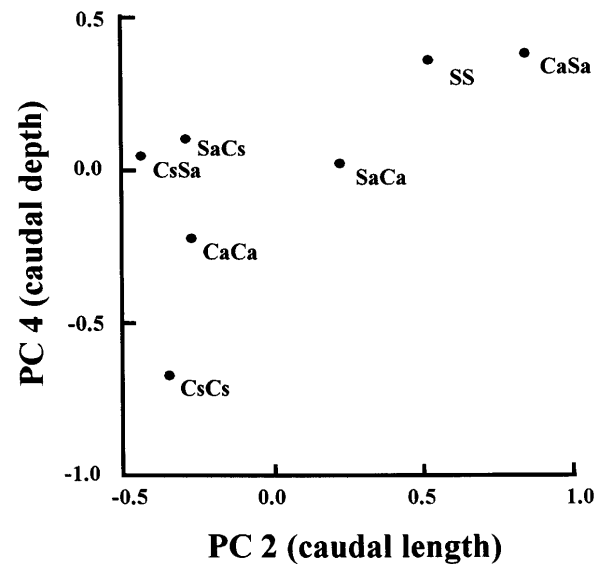
examined (PC1: $F = 17.76$, $df = 4, 177$, $p < 0.001$; PC2: $F = 12.84$, $df = 4, 177$, $p < 0.001$; PC4: $F = 5.46$, $df = 4, 177$, $p < 0.001$). However, Tukey's tests comparing fish within a rearing type only identified significant differences between wild cutthroat and steelhead in PC1 (head size) and between hatchery steelhead and Shelton cutthroat in PC1 (head size).

Discussion

Steelhead of hatchery origin were capable of higher critical swimming velocities than cutthroat of hatchery origin when reared in the same environment. The magnitude of this difference in mean critical swimming velocity ranged from 2.21 bl/s for steelhead and Shelton cutthroat to 1 bl/s for steelhead and Aberdeen cutthroat. These differences in swimming ability are consistent with differences in juvenile and adult habitat utilization and ocean migrations. Steelhead attain higher critical swimming velocities, are generally found in faster water in the streams as juveniles, and undergo more extensive migrations than cutthroat.

The morphological analysis showed that the differences in swimming ability were likely associated with heritable differences in body shape between the species and not differences

Fig. 4. Plot of mean principal components scores for PC2 (caudal length) and PC4 (caudal depth) derived from morphological analysis of cross types tested in swimming tunnel. In all fish types the dam is listed first and subscripts indicate origin of parent. C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton.



in hatchery practices. Cutthroat tended to have shorter and narrower caudal regions than steelhead, features poorly suited to sustained swimming at high velocities (Taylor and McPhail 1985; Taylor and Foote 1991). Furthermore, although the juvenile cutthroat and steelhead collected in the wild were generally more fusiform than the hatchery fish (possibly indicating increased swimming ability; Thomas and Donahoo 1977; Swain et al. 1991), the shape relationship between the species was consistent within rearing environments. Cutthroat from either hatchery or from the two streams had longer heads and narrower caudal regions than steelhead from the respective rearing environments. Therefore, the differences in swimming ability between cutthroat and steelhead in this study are relevant to at least some populations of wild fish. Wild cutthroat should exhibit lower critical swimming velocities than wild steelhead from the same region, but variation among populations exists (Thomas and Donahoo 1977; Tsuyuki and Willis-croft 1977; the present study).

The difference in swimming ability between the two stocks of cutthroat mirrored their morphology. Both sets of morphological measurements (modified truss and field identification) identified the Shelton cutthroat as having deeper and larger heads and deeper midbodies than any of the other fish types, including Aberdeen cutthroat. This was especially evident when head and body depth were compared with caudal region development. Shelton cutthroat had significantly poorer swimming ability and had a body shape least suited to sustained swimming. Although the difference was not significant, hybrids with Shelton parents (maternal or paternal) also tended to have poorer swimming ability than hybrids with Aberdeen parents. This would seem to indicate an influence on swimming ability associated with stock differences in body shape. It is possible that the increased swimming ability and more fusiform morphology of fish with Aberdeen cutthroat parents

Table 6. Rotated character component loadings (used to determine which characters comprise each principal component) and group mean component scores for the principal components analysis of morphological variation comparing hatchery and wild juvenile cutthroat and steelhead.

(A) Rotated component loadings					
Character	Fig. 2B No.	Rotated component loadings			
		PC1	PC2	PC3	PC4
Snout	1	0.878	-0.016	-0.058	0.100
Maxillary	2	0.867	-0.153	0.156	0.153
Head length	3	0.738	-0.128	0.327	0.346
Mouth width	Not shown	0.593	0.434	-0.501	0.044
Body depth	5	-0.021	0.868	-0.039	-0.153
Weight	Not shown	-0.124	0.817	-0.199	-0.284
Caudal depth	7	-0.045	0.746	0.166	0.208
Eye diameter	8	0.166	0.048	0.915	0.019
Mid-eye-hypural	9	-0.299	0.099	0.006	-0.896
% variation		28.38	24.64	14.36	12.29

(B) Mean component scores

Group	Mean component scores			
	PC1	PC2	PC3	PC4
C _a C _a	-0.355	0.198	-0.743	-0.329
C _s C _s	0.247	0.638	-1.111	-0.384
SS	-1.010	0.654	-0.218	-0.340
C _w C _w	0.416	-0.442	0.187	0.336
S _w S _w	-0.260	0.071	0.976	-0.240

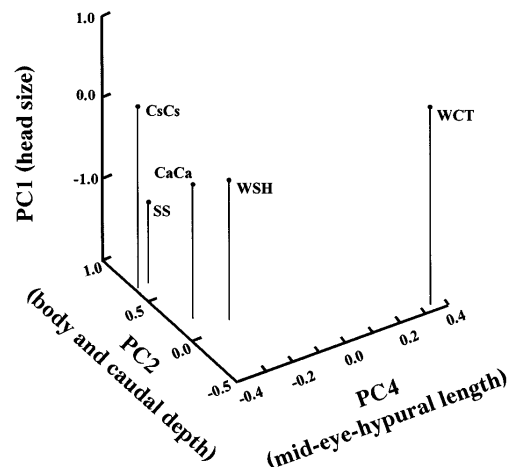
Note: The percentage of total variation explained by each component is given below the character component loadings. In all fish types the dam is listed first and subscripts indicate origin of parent. C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton; w, wild.

are due to steelhead genetic influence in the Aberdeen cutthroat stock.

Generally, cutthroat-steelhead hybrids had intermediate swimming abilities. The hybrid trout had mean critical swimming velocities closer to those of steelhead regardless of whether they had a maternal or paternal steelhead parent. There were no significant differences in swimming ability found among any of the hybrid types. The morphology of the hybrid trout was also generally intermediate to those of the parental species. Hybrids with Aberdeen cutthroat mothers were exceptional in both caudal length and depth, with mean component scores higher than all other fish types. These fish also outperformed the other hybrid types in the swimming tests, again indicating the strong association between caudal region development and swimming ability.

Brett (1965) indicated that although larger fish attain higher absolute sustained swimming speeds, they exhibit reduced swimming ability relative to their size (body lengths per second). Results from this study agree with Brett's results, except for steelhead; smaller fish performed better in the swimming tests in terms of relative speed (body lengths per second). This difference between steelhead and the other fish types tested is possibly due to differences in the morphology of the fish and how well suited they are to sustained swimming. Because of allometric growth in cutthroat and steelhead, the relative shape of the fish changes with fork length and therefore species differences become more pronounced in larger fish. For example,

Fig. 5. Plot of mean principal components scores for PC1 (head size), PC2 (body and caudal depth) and PC4 (mid-eye-hypural length) derived from morphological analysis of hatchery and wild cutthroat and steelhead. In all fish types the dam is listed first and subscripts indicate origin of parent. C, cutthroat; S, steelhead; a, Aberdeen; s, Shelton; w, wild.



it is very difficult to differentiate between juvenile cutthroat and steelhead less than 50 mm in length, yet it is relatively easy to distinguish them as adults. These allometric shape changes could be responsible for the fact that steelhead swimming ability improved with size whereas the other fish types became somewhat poorer swimmers with size. As a steelhead grows it rears in faster water and matures into a fish that undergoes extensive marine migrations, whereas cutthroat do not. It has been shown (Winans and Nishioka 1987) that there are changes in the shape of the caudal region of coho during the period of development associated with migration to salt water. However, further studies comparing the morphology of adult steelhead and cutthroat would need to be carried out to test this hypothesis.

The trend for fish tested earlier to perform better in the swimming tests is possibly due to the effects of a confined rearing environment on swimming ability. The fish continued to grow throughout the testing period and were kept in the same containers to minimize differences among types caused by environment.

The main purpose of this work was to assess the swimming ability and related morphology of cutthroat, steelhead, and hybrids in an attempt to address the relative performance and therefore the potential effects of occasional hybridization between coastal cutthroat and steelhead. It is unknown whether natural hybrid progeny assume a cutthroat, steelhead, or some intermediate life history, or to what age they survive. Previous work on cutthroat-steelhead hybrids (Campton 1981; Campton and Utter 1985) suggested that the potential for decreased fitness of the hybrids existed, because only young-of-the-year hybrids were detected. Furthermore, they were unable to positively determine if backcross or F₂ hybrids were present using protein allozymes. Swimming ability has fitness consequences both in terms of vulnerability to predation (Bams 1967) and in relation to the energetic demands of migration (Tsuyuki and Willisroft 1977; Taylor and McPhail 1985). Therefore, comparison of swimming ability could yield insights into the rela-

tive viability of the hybrids and parental cutthroat and steelhead. Hybrid trout were intermediate in both swimming ability and morphology, putting them at a competitive disadvantage when compared with steelhead in fast water. However, all of the hybrid types tested demonstrated superior swimming ability and morphology better suited to sustained swimming than cutthroat. Habitat use by juvenile cutthroat is generally thought to be dictated by competitive displacement by coho and steelhead and a lack of morphological adaptation to either fast or slow water (Bisson 1988). In streams containing all three salmonids, cutthroat often occupy less desirable, intermediate habitat. However, in the absence of coho, cutthroat move into deeper, slower water (Johnston 1982; Bisson et al. 1988; Pauley et al. 1989). Consequently, the relative competitiveness of hybrid and cutthroat trout based on swimming ability and related morphology could be influenced by the presence of other salmonids. In a stream without coho, hybrids might occupy the intermediate habitat, cutthroat might reside in pools, and steelhead might live in riffles. In this case, the hybrids might be at a competitive disadvantage, analogous to cutthroat in a stream containing coho and steelhead. However, in a stream containing coho, the competitive outcome is less clear. If cutthroat and hybrids competed for the intermediate habitat, the superior swimming ability of the hybrids might put them at a competitive advantage. Therefore, this comparison of relative swimming ability does not demonstrate hybrid depression. There is the potential for a competitive advantage when cutthroat-steelhead hybrids are compared with cutthroat. However, swimming ability is just one measure of performance and further studies comparing other traits, such as aggression and habitat preference, will be important in predicting the viability of hybrids relative to the parental species.

In summary, juvenile steelhead achieved higher critical swimming velocities than cutthroat and hybrids were intermediate in both swimming performance and morphology. Depending on the life history adopted by hybrids, there is the potential for a competitive advantage over cutthroat. This advantage of the F_1 progeny could then contribute to introgression between coastal cutthroat and steelhead.

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